Review retrospective: *moment-normalisation — partition derivatives are Gibbs moments, and two loose glosses*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicMomentNormalisation.lean`, which divides the partition-derivative result $Z^{(n)}(0) = \int(-(tx))^n e^{-tL}$ by $Z(0) = \int e^{-tL}$ to identify it with the unperturbed Gibbs expectation $\langle(-(tx))^n\rangle_0 =$ `gibbsExp` — tying the laplace partition-derivative arc to `Threepoint.gibbsExp`. Result: **a clean theorem (statement and proof sound), no edits**, plus two report-only prose imprecisions in the closing docstring gloss.

1 Setting

The laplace anharmonic arc computes iterated h-derivatives of the perturbed partition $Z(h)$. This file is the normalisation step: $Z^{(n)}(0)/Z(0)$ is the n-th raw Gibbs moment of the derivative-insertion observable, expressed through the cross-seabed `Threepoint.gibbsExp` used by the FDT capstone.

2 What was reviewed

The single public theorem `iteratedDeriv_partition_div_eq_gibbsExp`.

3 Fidelity / soundness findings

Theorem and proof sound; GPT found no theorem-level mismatch. After `rw [iteratedDeriv_weightedPartition_zero …]` the numerator is $\int(-(tx))^n e^{-tL}$; at $h = 0$, `gibbsExp (volume) L id t 0` $(x \mapsto (-(tx))^n)$ is the same quotient with denominator $\int e^{-tL}$, and `unfold` + `simp [pow_zero, one_mul]` reduces `gibbsExp`'s denominator ($\int(-(tx))^0 e^{-tL}$) to $\int e^{-tL}$, matching $Z(0)$. The hypotheses ($\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$) are the anharmonic well-definedness regime, consumed by the cited derivative theorem.

Two report-only prose imprecisions (closing gloss).

- The closing sentence calls this "the n-th derivative of the *log-partition's* first integral". The theorem is $Z^{(n)}/Z$ (raw moments); $(\log Z)^{(n)}$ for $n \geq 2$ are *cumulants*, not raw moments. "The log-partition's first integral" defensibly means the integral Z *inside* $\log Z$ (whose normalised derivative is the raw moment), but it is mis-readable.
- "the n-th raw moment of the *perturbation observable*": the formal observable is $(-(tx))^n = (-tA)^n$ — moments of the derivative-insertion $-tA$, not the unscaled $A^n = x^n$.

Both are *report-only*: the docstring's *explicit formulas* two lines above are correct and present ($Z^{(n)}(0)/Z(0) = \langle(-(tx))^n\rangle_0$, with the observable written as $(-(tx))^n$), so the body self-corrects the loose closing prose. This is the defensible-gloss class (cf. the patterning naming and stale-title notes), not the objective wrong-value class of the landed $\kappa_3 = 2$ / $Z = s\int e^{-sK}$ fixes; the precise reword is the author's call. (A third item GPT raised — a missing γ in the potential — was an artefact of the reviewer's prompt summary, not the file, which correctly uses the 3-parameter `anharmonicPotential`.)

4 Simplifications applied

None. GPT's (B) findings (collapse `unfold;simp` to one `simp`; drop the in-proof comment) are golf, declined; `open MeasureTheory` is load-bearing (`Measure`/`volume` in the statement), and the γ/discriminant/t hypotheses are consumed by the cited derivative theorem (not redundant).

5 Disagreements and open questions

None; GPT and I agree the core is sound and the glosses are prose-only.

6 Lessons

- **When an explicit formula and a prose gloss sit side by side, the gloss's looseness is report-only.** The docstring states the exact identity $Z^{(n)}/Z = \langle(-(tx))^n\rangle$ explicitly, then paraphrases it as "the log-partition…raw moment of the perturbation observable". The paraphrase drops the $-t$ scaling and risks a $\log Z$/cumulant misreading, but the adjacent exact formula prevents any actual confusion — so it is a clarity nit for the author, not a correctness fix. The reviewer's job is to *flag* the paraphrase, not to rewrite a self-clarified docstring.

7 Follow-ups

- Author (optional): reword the closing gloss — "the partition Z (the integral inside $\log Z$)" and "the n-th raw moment of the derivative-insertion $-tA$" — for precision.
- The laplace OneD anharmonic arc has many never-reviewed siblings (`Anharmonic{FDT,SecondCumulant, ThirdCumulant,CumulantsLogPartition,…}`) — a fresh-context review arc for the next session.